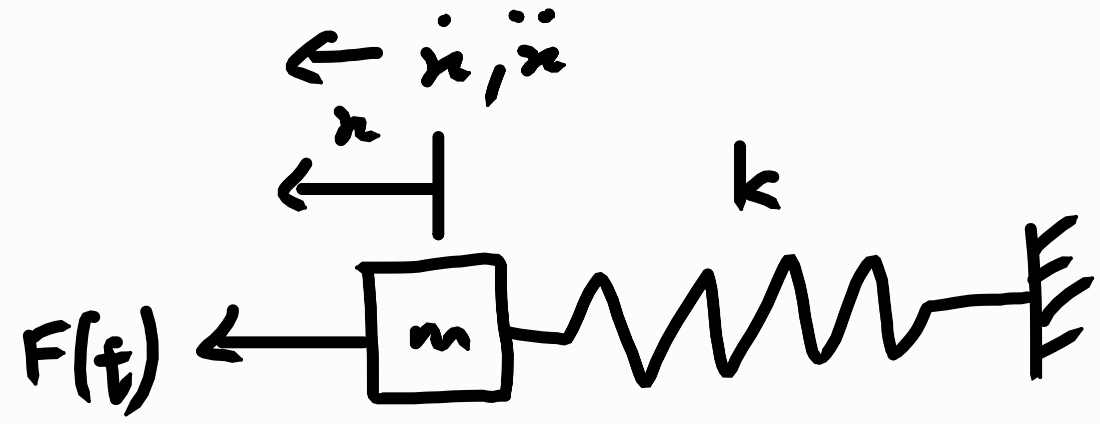
1)

$$F(t) = 5\sin\omega t$$

$$\omega = 2\pi\left(\frac{60}{60}\right)$$

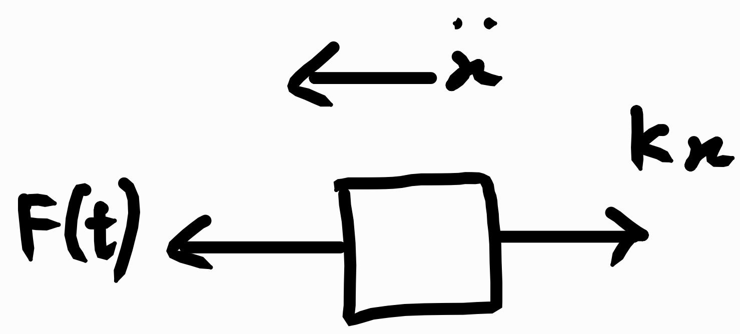
$$= 2\pi \text{ rad s}^{-1}$$



$$m = m_{eff} + M_1$$

$$= 5 + 5$$

$$= 10 \text{ kg}$$



$$m\ddot{x} = \sum F_x$$

$$m\ddot{x} = F(t) - kx$$

$$m\ddot{x} = F_0 \sin\omega t - kx$$

$$m\ddot{x} + kx = F_0 \sin\omega t$$

Assuming the steady-state solution is in the form:

$$x(t) = X\sin(\omega t - \phi) = X\sin\omega t$$

$\because \phi = 0$ as damping is 0

$$X = \frac{\frac{F_0}{k}}{1-\left(\frac{\omega}{\omega_n}\right)^2} = \frac{F_0}{k - \frac{k\omega^2}{\frac{k}{m}}} = \frac{F_0}{k - \omega^2 m}$$

$$= \frac{5}{1000 - (2\pi)^2(10)}$$

$$= 8.261515648 \times 10^{-3}$$

$$\approx 8.26 \text{ mm}$$

2) $\omega = 2\pi\left(\frac{950}{60}\right)$

$= \frac{95}{3}\pi$

For one set of rubber pads:

$k_1 = k$

$\omega_{n_1} = \sqrt{\frac{k}{m}} = 2\pi\left(\frac{500}{60}\right) = \frac{50}{3}\pi$

For two sets of rubber pads:

$\frac{1}{k_2} = \frac{1}{k} + \frac{1}{k}$

$\frac{1}{k_2} = \frac{2\cancel{k}}{k^{\cancel{2}}}$

$k_2 = \frac{k}{2}$

$\omega_{n_2} = \sqrt{\frac{\frac{1}{2}k}{m}} = \frac{1}{\sqrt{2}}\sqrt{\frac{k}{m}}$

$= \frac{1}{\sqrt{2}}\omega_{n_1}$

$= \frac{1}{\sqrt{2}}\left(\frac{50}{3}\pi\right)$

$= \frac{25\sqrt{2}}{3}\pi$

2) For both situations:

$$m\ddot{x} + kx = F_0 \sin \omega t$$

$$x(t) = X \sin \omega t$$

$$X = \frac{F_0}{k - \omega^2 m}$$

$$\frac{X_2}{X_1} = \frac{\cancel{F_0}}{k_1 - \omega^2 m} \times \frac{k_2 - \omega^2 m}{\cancel{F_0}}$$

$$X_2 = \frac{k_2 - \omega^2 m}{k_1 - \omega^2 m} X_1$$

$$= \frac{\frac{k_2}{m} - \omega^2}{\frac{k_1}{m} - \omega^2} X_1$$

$$= \frac{\omega_{n_2}^2 - \omega^2}{\omega_{n_1}^2 - \omega^2} X_1$$

$$= \frac{\left(\frac{25\sqrt{2}}{3}\pi\right)^2 - \left(\frac{95}{3}\pi\right)^2}{\left(\frac{50}{3}\pi\right)^2 - \left(\frac{95}{3}\pi\right)^2} \times 3.8$$

$$= 3.189067524$$

$$\approx 3.2 \text{ mm}$$

3) Equation of motion:

$$m\ddot{x} + c\dot{x} + kx = F_0 \sin \omega t$$

$m = 1.95$, $F_0 = 24.46$ $X_{res} = 1.27\,cm$

$$\omega = \frac{2\pi}{0.2} = 10\pi \text{ rads}^{-1}$$

Steady-state solution is of the form:

$$x(t) = X \sin(\omega t - \phi)$$

Substituting the above into the equation of motion:

$$X = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$$

At resonant frequency:

$$X_{res} = \frac{F_0}{\sqrt{(k - m\omega_n^2)^2 + (c\omega_n)^2}}$$

$$X_{res} = \frac{F_0}{\sqrt{\cancel{(k - m\frac{k}{m})^2}^{\,0} + (c\omega_n)^2}}$$

$$X_{res} = \frac{F_0}{c\omega_n}$$

3) $\therefore c = \dfrac{F_0}{x_{res}\omega_n}$

$= \dfrac{24.46}{1.27\times10^{-2}\times10\pi}$

$= 61.3059828$

$\approx 61.31 \text{ Ns m}^{-1}$

$\xi = \dfrac{c}{c_c} = \dfrac{c}{2m\omega_n} = \dfrac{61.31}{2(1.95)(10\pi)}$

$= 0.5003666771$

≈ 0.5

4) $M = 25$

$$\frac{X_0}{X_4} = \left(\frac{X_0}{X_1}\right)^4 = \frac{16}{1}$$

$$\frac{X_0}{X_1} = 2$$

$$\delta = \ln\left(\frac{X_0}{X_1}\right) = \ln(2) \approx 0.693$$

$$\zeta = \sqrt{\frac{\delta^2}{4\pi^2 + \delta^2}}$$

$$= \sqrt{\frac{0.693^2}{4\pi^2 + 0.693^2}}$$

$$= 0.109652581 \approx 0.1097$$

Equation of motion:

$$M\ddot{x} + c\dot{x} + kx = me\omega^2 \sin\omega t$$

Steady-state solution:

$$x(t) = X \sin\omega t$$

Substituting into the equation of motion:

$$X = \frac{me\omega^2}{\sqrt{(k - M\omega^2)^2 + (c\omega)^2}}$$

4) $$\frac{MX}{me} = \frac{\left(\frac{w}{w_n}\right)^2}{\sqrt{\left(1-\frac{w^2}{w_n^2}\right)^2 + \left(\frac{2\zeta w}{w_n}\right)^2}}$$

At resonant frequency, $w = w_n$,

$$\frac{MX}{me} = \frac{\cancel{\left(\frac{w_n}{w_n}\right)^2}\ 1}{\cancel{\sqrt{}}\underbrace{\cancel{\left(1-\frac{w_n^2}{w_n^2}\right)^2}}_{0} + \left(\frac{2\zeta \cancel{w_n}}{\cancel{w_n}}\right)^{\cancel{2}}}$$

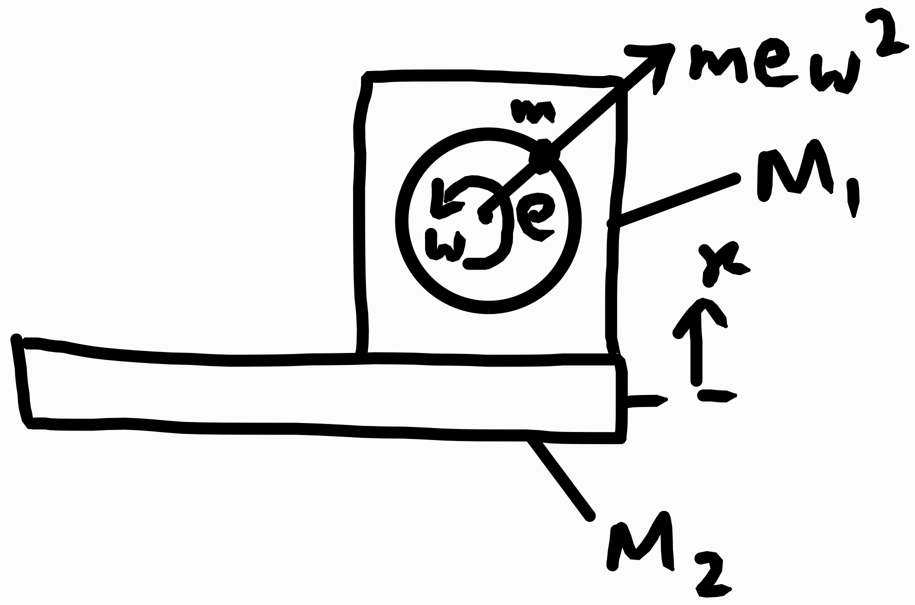
$$= \frac{1}{2\zeta}$$

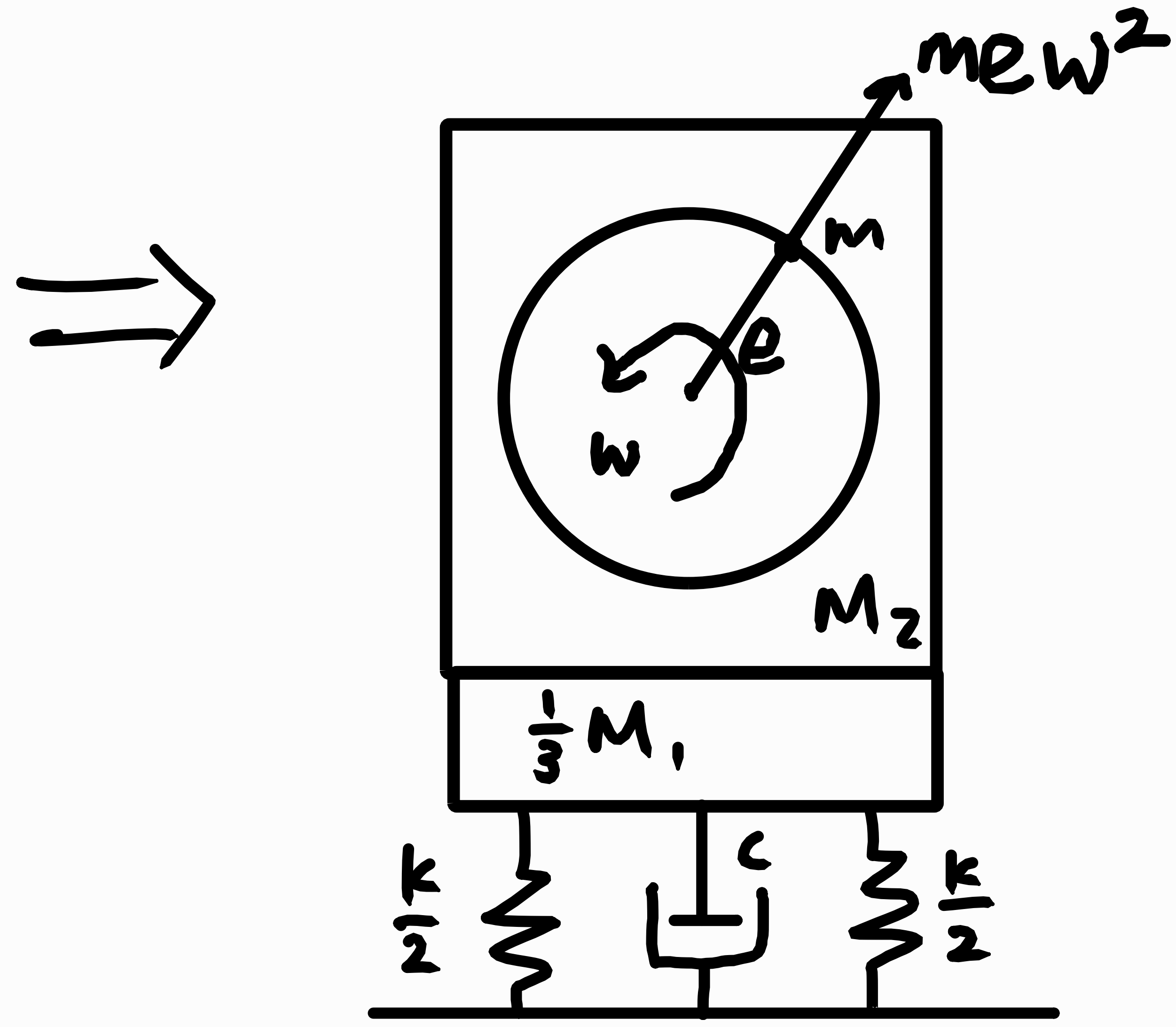
$$= \frac{1}{2(0.01097)}$$

$$= 4.559856188$$

$$\approx 4.56$$

5)





$M = \frac{1}{3}M_1 + M_2 + m$

$= \frac{1}{3}(60) + 20 + 0.5$

$= 40.5$

$\zeta = \frac{c}{c_c} = 0.01$

$c = 0.01 c_c = 0.01(2\sqrt{kM})$

$= 0.01(2\sqrt{(100 \times 10^3)(40.5)})$

$= 40.24922359$

$\approx 40.25 \text{ Ns m}^{-1}$

$w = 2\pi\left(\frac{1500}{60}\right)$

$= 50\pi \text{ rad s}^{-1}$

5.1) Vibration amplitude:

$$X = \frac{mew^2}{\sqrt{(k-Mw^2)^2+(cw)^2}}$$

$$= \frac{0.5(15\times10^{-2})(50\pi)^2}{\sqrt{(100\times10^3-40.5(50\pi)^2)^2+(40.25\times50\pi)^2}}$$

$$= 2.05772364\times10^{-3}\ m$$

$$\approx 2.06mm$$

5.2) $w_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{100\times10^3}{40.5}}$

$$= 49.6903995$$

$$\approx 49.69 rads^{-1}$$

$$X_{res} = \frac{mew_n^2}{\sqrt{(k-Mw_n^2)^2+(cw_n)^2}}$$

$$= \frac{0.5(15\times10^{-2})(49.69)^2}{\sqrt{(100\times10^3-40.5(49.69)^2)^2+(40.25\times49.69)^2}}$$

$$= \frac{5}{54}$$

$$\approx 92.6\ mm$$